



Establishing linear equivalences

Teacher resource

Question 1

Which one of the following expressions is equivalent to $5b + 13$?

- ☐ $2b \times 3b + 13$
- ☐ $5 + b + 13$
- ☐ $18b$
- ☐ $b + 18 + 4b - 5$

Answer: D

Skill: Students simplify a linear expression.

Additional questions

1. If $b = 1$, what is the value of each of the five expressions?
Which expressions have the same numerical value when $b = 1$?
2. If $b = -2$, what is the value of each of the five expressions?
Which expressions have the same numerical value when $b = -2$?

Question 2

Which one of the following expressions is equivalent to $3(4m + 1)$?

- ☐ $7m + 1$
- ☐ $12m + 1$
- ☐ $12m + 3$
- ☐ $15m$

Answer: C

Skill: Students expand brackets in a linear expression.

Additional questions

1. If $m = 1$, what is the value of each of the five expressions?
Which expressions have the same value when $m = 1$?
2. If $m = -2$, what is the value of each of the five expressions?
Which expressions have the same value when $m = -2$?

Question 3

Four of the five expressions below are equivalent.
Choose the expression which is not equivalent to the others.

- ☐ $2a \times 2 - 8$
- ☐ $4a - 8$
- ☐ $2(2a - 4)$
- ☐ $4(a - 2)$
- ☐ $-2(-2a - 4)$

Answer: E

Skill: Students identify equivalent linear expressions.

Additional questions

1. If $a = 1$, what is the value of each of the above expressions?
Which expressions have the same value when $a = 1$?
2. If $a = 0$, what is the value of each of the above expressions?
Which expressions have the same value when $a = 0$?
3. If $a = -2$, what is the value of each of the above expressions?
Which expressions have the same value when $a = -2$?

Question 4

Which of these algebraic expressions are linear?

- ☐ $2x(1 - x) + 2 + x^2$
- ☐ $2x(1 - x) + 2x^2$
- ☐ $\frac{2x}{3}(x + 3) - 2x(x - \frac{1}{3})$
- ☐ $\frac{2x}{3}(x + 3) - x(\frac{2}{3}x - 2)$

Answer: B and D

Skill: Students simplify and identify linear expressions

Additional questions

1. Graphs could be drawn for each of the two linear expressions. For each of these graphs, predict the x-intercept, y-intercept and gradient.
2. Construct a table of values for $-5 \leq x \leq 5$ for each of the other two expressions.
Use these tables to draw graphs of these two expressions.

Student activity: Establishing linear equivalences

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- ☐ $2x(1 - x) + 2x^2$
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- ☐ $\frac{2x}{3}(x + 3) - x(\frac{2}{3}x - 2)$